suitable nesting sites can all be limiting factors. For example, the carrying capacity for bats may be high in a habitat with abundant flying insects and roosting sites but lower where there is abundant food but fewer suitable shelters.

Crowding and resource limitation can have a profound effect on population growth rate. If individuals cannot obtain sufficient resources to reproduce, then the per capita birth rate will decline. Similarly, if starvation or disease increases with density, then the per capita death rate may increase. Falling per capita birth rates or rising per capita death rates will cause the per capita rate of population growth to drop, a very different situation from the constant per capita growth rate (r) seen in a population that is growing exponentially.

The Logistic Growth Model

We can modify our mathematical model so that the per capita population growth rate decreases as N increases. In the **logistic population growth** model, the per capita rate of population growth approaches zero as the population size nears the carrying capacity (K).

To construct the logistic model, we start with the exponential population growth model and add an expression that reduces the per capita rate of population growth as N increases. If the carrying capacity is K, then $K - N$ is the number of additional individuals the environment can support, and $(K - N)/K$ is the fraction of K that is still available for population growth. By multiplying the exponential rate of population growth rN by $(K - N)/K$, we modify the change in population size as N increases:

$$\frac{dN}{dt} = rN\frac{(K - N)}{K}$$

When N is small compared to K, the term $(K - N)/K$ is close to 1. In this scenario, the per capita rate of population growth, $r[(K - N)/K]$, will be close to (but slightly less than) r, the intrinsic rate of increase seen in exponential population growth. But when N is large and resources are limiting, then $(K - N)/K$ is close to 0, and the per capita population growth rate is small. When N equals K, the population stops growing. **Table 53.2** shows calculations of population growth rate for a hypothetical population growing according to the logistic model, with $r = 1.0$ per individual per year. Notice that the overall population growth rate is highest, +375 individuals per year, when the population size is 750, or half the carrying capacity. At a population size of 750, the per capita population growth rate remains relatively high (one-half the value of r), and there are more reproducing individuals (N) in the population than at lower population sizes.

As shown in **Figure 53.9**, the logistic model of population growth produces a sigmoid (S-shaped) growth curve when N is plotted over time (the red line). New individuals are added to the population most rapidly at intermediate population sizes, when there is not only a breeding population of substantial size, but also lots of available space and other resources in the environment. The number of individuals added to the population decreases dramatically as N approaches K. As a result, the population growth rate (dN/dt) also decreases as N approaches K.

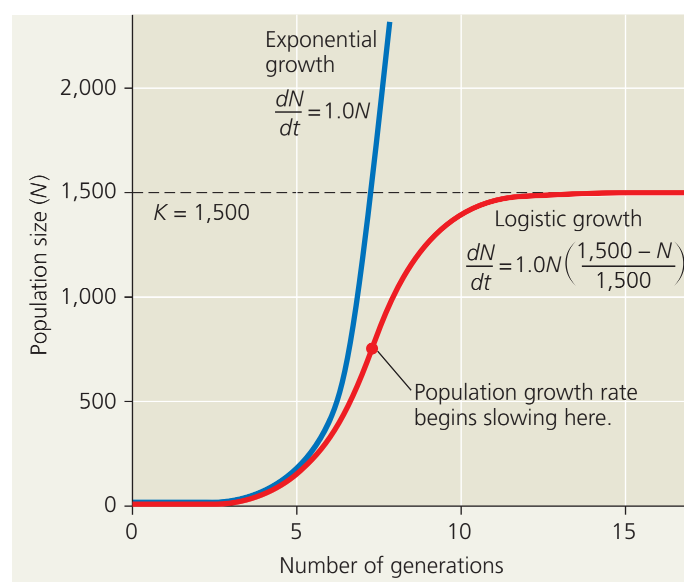
Note that we haven't said anything yet about *why* the population growth rate decreases as N approaches K. For a population's growth rate to decrease, the birth rate must decrease,

Table 53.2 Logistic Growth of a Hypothetical Population ($K = 1{,}500$)

Population Size (N)	Intrinsic Rate of Increase (r)	$\frac{K-N}{K}$	Per Capita Population Growth Rate, $r\frac{(K-N)}{K}$	Population Growth Rate,* $rN\frac{(K-N)}{K}$
25	1.0	0.983	0.983	+25
100	1.0	0.933	0.933	+93
250	1.0	0.833	0.833	+208
500	1.0	0.667	0.667	+333
750	1.0	0.500	0.500	+375
1,000	1.0	0.333	0.333	+333
1,500	1.0	0.000	0.000	0

*Rounded to the nearest whole number.

▼ **Figure 53.9 Population growth predicted by the logistic model.** The rate of population growth decreases as population size (N) approaches the carrying capacity (K) of the environment. The red line shows logistic growth in a population where $r = 1.0$ and $K = 1{,}500$ individuals. For comparison, the blue line illustrates a population continuing to grow exponentially with the same r.



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